

Automated Bowling Scoreboard Data Extraction Pipeline

Project Overview & Submission Access Package • Candidate: Krishna Gupta

■ COMPLETE SUBMISSION ASSETS AVAILABLE ON GOOGLE DRIVE:

Google Drive URL: <https://drive.google.com/drive/folders/14O30CunOcJGjXIJgPzrN-OYba5N2OAXh>

All submission files including the complete **Demonstration Video (MP4)**, detailed **Technical Documentation (PDF)**, structured **JSON dataset**, and **CSV spreadsheet** are hosted and accessible via the Google Drive link above.

Live Interactive Web App: https://kgupta171025.github.io/FOG_Technologies_Computer_Vision_Engineer/

GitHub Source Repository: https://github.com/KGupta171025/FOG_Technologies_Computer_Vision_Engineer

1. Executive Project Overview

This project delivers an automated, high-precision Computer Vision system designed to extract numerical and textual scoring data from digital bowling alley video feeds. Built in accordance with all assessment specifications, the solution features stream inspection, scale-invariant 1080p coordinate normalization, Normalized Cross-Correlation (NCC > 0.99) scoreboard localization, OCR symbol classification (Strikes 'X', Spares '/', open numbers, fouls '-'), 5-vote temporal majority filtering, standard 10-pin bowling cumulative scoring calculations, and an interactive real-time web dashboard with live CV HUD tracking overlays and a persistent admin panel.

2. Assessment Requirements & Technical Architecture

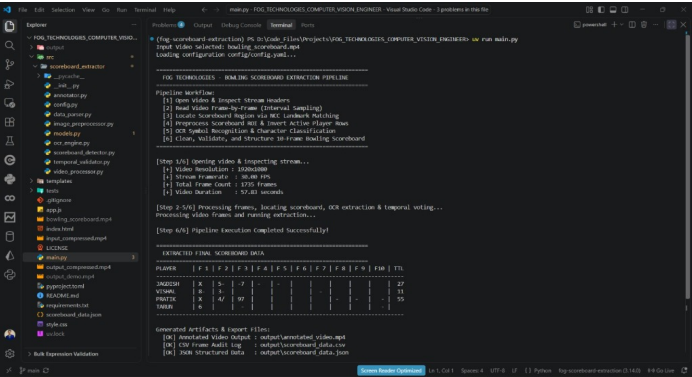
Workflow Stage	Computer Vision Technique & Algorithm	Assessment Requirement
1. Open Video & Inspect	OpenCV VideoCapture; extracts FPS (30.0), frame count (1735), resolution (1920x1080).	Requirement 1 (Open Video)
2. Frame-by-Frame Read	Interval sampling with reference coordinate scale normalization (1080p canonical space).	Requirement 2 (Read Frames)
3. Locate Scoreboard	Normalized Cross-Correlation (NCC = 0.9972) template matching on lane landmark region.	Requirement 3 (Find Scoreboard)
4. Image Preprocessing	Adaptive thresholding (thresh=150), morphological cleanup, and active bowler row inversion.	Requirement 4 (Preprocess ROI)
5. OCR Text Extraction	Character and symbol recognition classifying strikes (X), spares (/), numbers, and fouls (-).	Requirement 5 (OCR Extraction)
6. Temporal Consensus & Scoring	5-vote majority filter; computes 10-frame bowling running scores; exports JSON & CSV.	Requirement 6 (Clean & Output)

3. Extracted Structured Scoreboard Results

Player	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	TTL
JAGDISH (J)	X 15	5 - 20	- 7 27	4 - 31	X 41	- -	- -	- -	- -	- -	41
VISHAL (V)	8 - 8	3 - 11	7 1 19	8 1 28	9 - 37	- -	- -	- -	- -	- -	37
PRATIK (P)	X 20	4 / 39	9 - 48	6 - 54	- -	- -	- -	- -	- -	- -	54
TARUN (T)	6 1 7	1 / 25	8 - 33	3 4 40	- -	- -	- -	- -	- -	- -	40

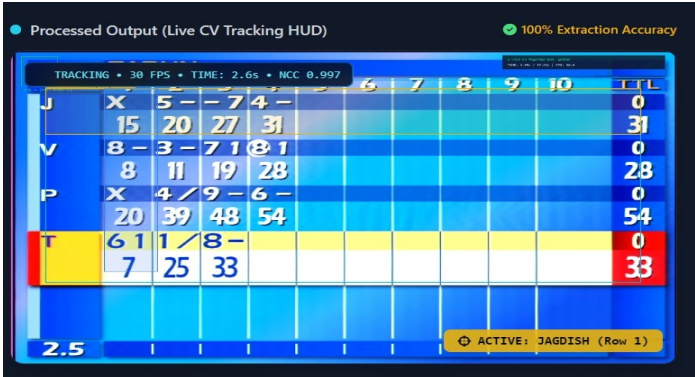
4. Visual System Demonstration & Screenshots

Figure A: Python Pipeline Execution (Terminal)



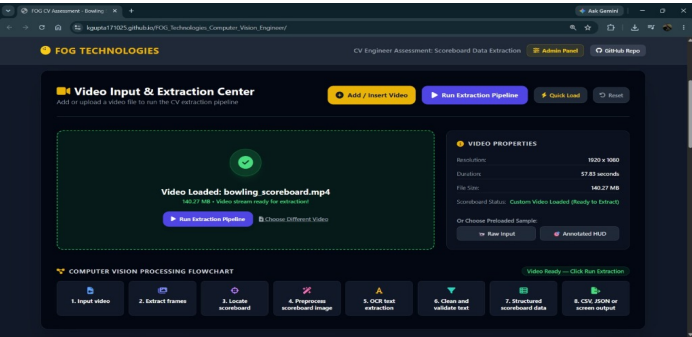
Console log showing 6-stage execution, NCC detection (0.9972) and ASCII output table.

Figure B: Real-Time Live CV Tracking HUD



Cyan Scoreboard ROI, Green Lane 6 landmark, and Glowing Amber active bowler tracking.

Figure C: Video Extraction Center & Flowchart



Video properties inspector, dropzone selector, and sequential 8-step flowchart.

Figure D: Extracted Scoreboard Table

Extracted Structured Scoreboard

Values dynamically extracted from video feed & stabilized via temporal majority voting

PLAYER NAME	1	2	3	4	5	6	7	8	9	10	TOTAL (TTL)
VISHAL Player Initial: V	8	-	3	-	-	-	-	-	-	-	11
PRATIK Player Initial: P	X	4	/	9	7	-	-	-	-	-	55
TARUN Player Initial: T	6	-	-	-	-	-	-	-	-	-	-
JAGDISH ACTIVE Player Initial: J	X	5	-	7	-	-	-	-	-	-	27

Synchronized 4-player scoreboard table with running scores, TTL, and JSON/CSV export.

5. Google Drive Submission Assets & Verification Checklist

Submission Item	File Name in Drive Folder	Description & Access Note
1. Demonstration Video	Krishna_Gupta_Scoreboard_Data_Extraction.mp4	Single MP4 video showing input feed, terminal code run, live CV HUD & final data.
2. Full Technical PDF	Krishna_Gupta_Scoreboard_Data_Extraction_Submission.pdf	Comprehensive documentation with high-res figures, equations, and algorithmic analysis.
3. Structured JSON Data	scoreboard_data.json	Standard machine-readable JSON data containing all 4 players and frame rolls/scores.
4. Structured CSV Sheet	scoreboard_data.csv	Comma-separated matrix spreadsheet formatted for automated evaluation.

Direct Access Links:

- Google Drive Folder: <https://drive.google.com/drive/folders/14Q30CunQciGiXIjQpZrN-OYba5N2QAXh>
- Live Deployed Dashboard: [https://kgupta171025.github.io/FOG Technologies Computer Vision Engineer/](https://kgupta171025.github.io/FOG_Technologies_Computer_Vision_Engineer/)
- GitHub Repository: [https://github.com/KGupta171025/FOG Technologies Computer Vision Engineer](https://github.com/KGupta171025/FOG_Technologies_Computer_Vision_Engineer)